

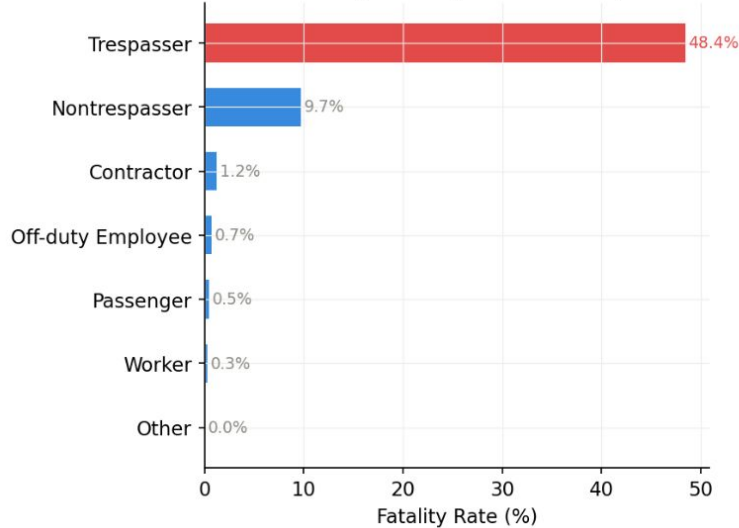
Duke Data Mini#3

How Behavioral and Spatial Patterns Shape Railroad Fatalities

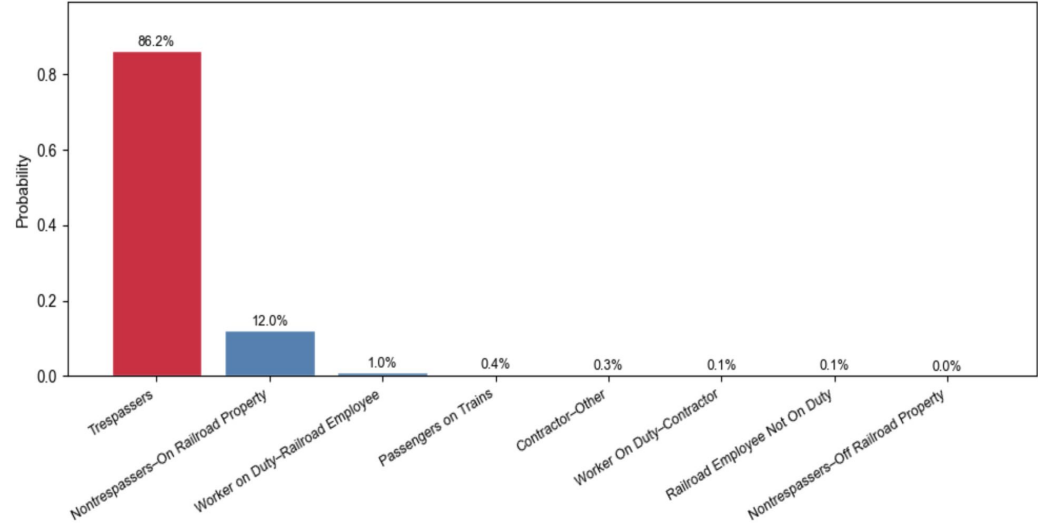
Eric Hu, Ruitong Liu, Mia Zhou

Problem: Who is dying on America's railroads?

Fatality Rate by Person Group



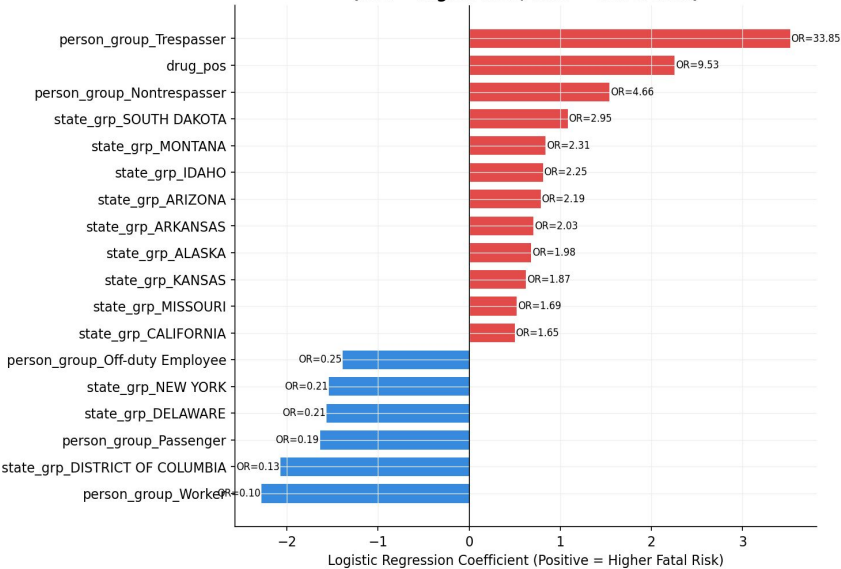
Fatal Incidents by Type of Person



Once injured, a **trespasser** has a **48.4%** chance of not surviving
Trespassers account for **86.2%** of all railroad fatalities

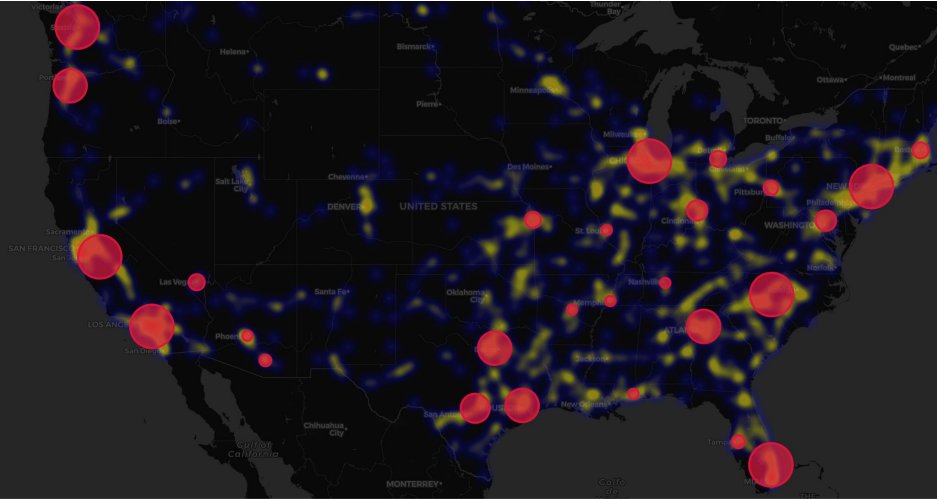
Delve Into Data

Fatal Risk Predictors - Logistic Regression Coefficients
(Red = Higher Risk, Blue = Lower Risk)

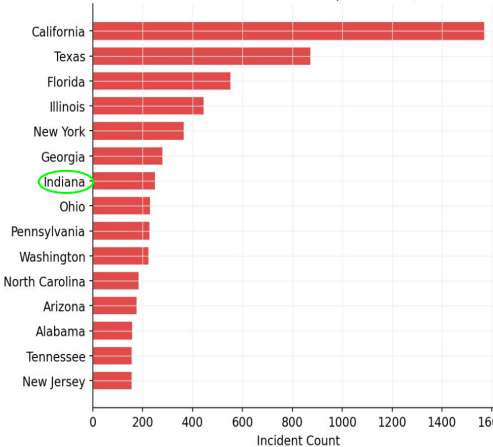


We built a logistic regression model using person type, state, age, time of day, and substance test results, which correctly identified fatal vs non-fatal incidents 92.4% of the time (AUC = 0.924).
33.85 Odds (being a trespasser): trespassers are nearly 34 times more likely to die than the baseline, holding others constant.
9.53 Odds(Positive drug test)

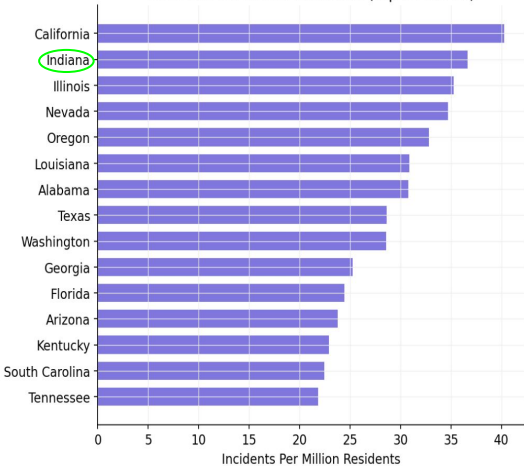
Trespasser Fatality Hotspots Across the U.S.



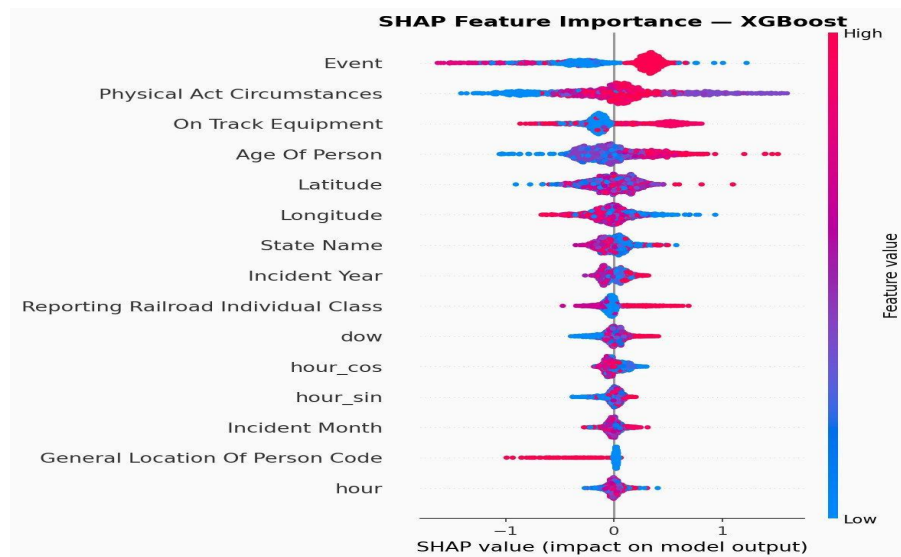
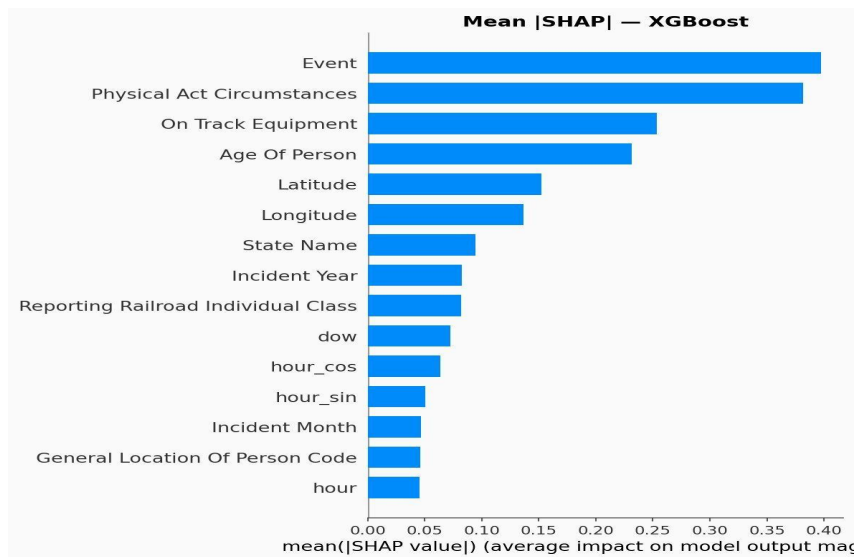
Raw Incident Count (Top 15 States)



Incidents Per Million Residents (Top 15 States)



PART A: WHAT PREDICTS TRESPASSER DEATH?



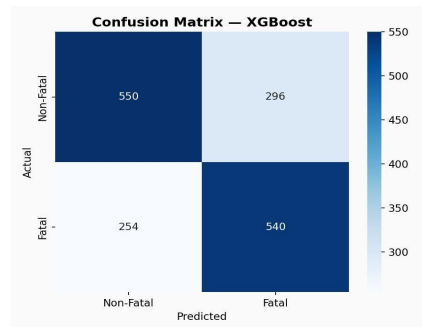
Key Findings

Event type is the #1 predictor: 'struck by train' is near-certain death

Physical act (laying vs walking) controls escape probability

Freight trains are far more lethal than passenger trains

Age 50+ sharply increases fatality risk under same conditions



Model Performance

XGBoost Best

AUC: 0.727

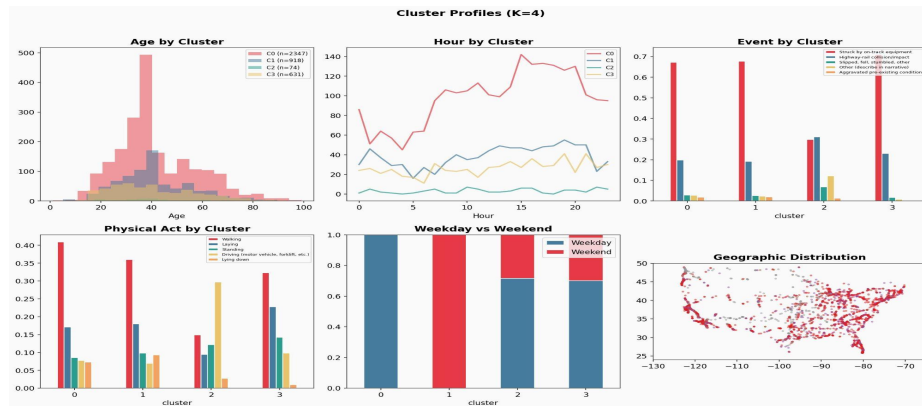
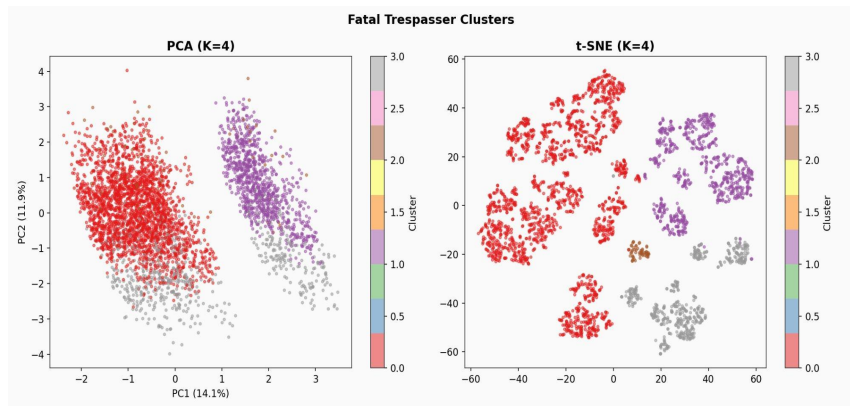
F1: 0.663

CV: 0.737 $\pm$ 0.007

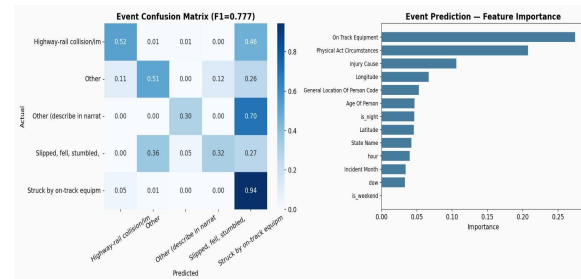
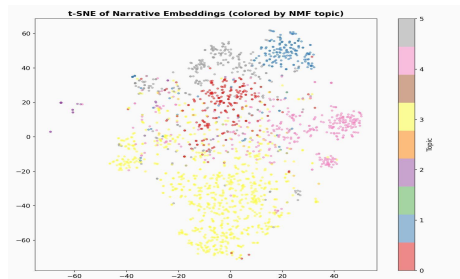
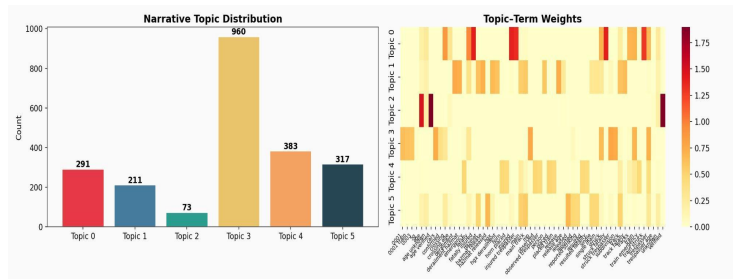
Recall (Fatal): 68%

Recall (Non-Fatal): 65%

PART B: FATAL TRESPASSER PROFILING



4 Archetypes: C0 Track-walkers struck by train (59%) | C1 Grade-crossing drivers (23%) | C2 Non-typical/weekend (2%) | C3 Western corridor mixed (16%)



NLP independently validates clustering — structured data and narrative text tell the same story | Event classification F1 = 0.777